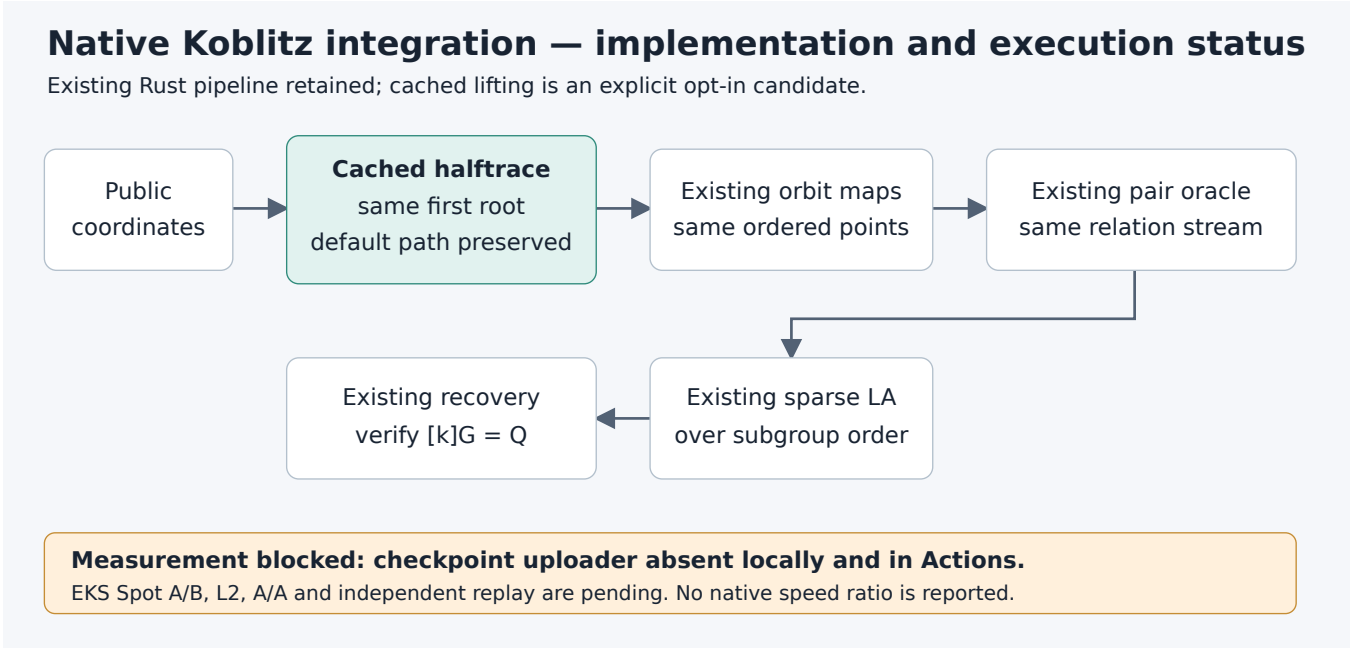
Native Koblitz cached lifting

2026-10-30 · Engineering implementation candidate · Generic decomposition track

Measurement blocked. Local checkpoint upload lacks an approved uploader. Actions run 41082358257 also failed: the upload secret is not configured. All build/experiment jobs were skipped. No native speed ratio is claimed.



Implementation

Lazy, curve-local byte tables evaluate the existing odd-degree halftrace. The explicit factor-base parameter `halftrace_tables=1` preserves the first root, ordered points, signed orbit maps and existing relation/LA implementation. Default recipes retain the original point-lifting path. Initialized immutable tables are shared by clones. No field-wide table or new Python path is introduced.

Verification and accounting

The original release suite completed: **5447 passed, zero failed, 123 existing ignored**. Final-source cached/ordered tests, both native E2E examples and strict library Clippy pass separately. The suite predates only the documented parity lint cleanup. Later main merges preserve the feature files but expose an existing syntax error in ecbench/methods.rs; their repeated check is blocked. Exact source scope and failures are retained under `development/`. Test times are not benchmarks.

The frozen comparison has eight independent one-target workloads, the same base and seed in both arms, strong-rho and IC A/A controls, six alternating measured rounds and one warmup. Its canonical protocol is prospectively frozen. Of 280 planned scientific executions, zero were attempted. EKS Spot L2 and independent replay are mandatory; cold cache construction is included.

Limits and sources

The degree-31 invariant subspace has ord_31(2)=5; it does not discharge the degree-83 confidence gate or establish transfer to ECC2K-130. Existing coarse Artin–Schreier solve counters alone do not price the cache delta. External arithmetic comparisons have not been measured. The scoreboard and leaderboard remain unchanged because no admitted performance result exists.

Saved baseline source: `3b34d504bc13ce668e86d8e68cdf57e3fbaba89a`. Scientific source pin: `7c58661935b79153ff74c1208d0592aa56ed04a1`. The primary binary point-recovery contract is SEC 1 v2.0, section 2.3.4. Existing `src/binary_ecc/f2m.rs` supplies the native trace/halftrace precedent. The study's `README.md`, `PROTOCOL.md`, `protocol.json` and `BASELINE.txt` retain source paths and evidence links. Same-binary flag comparison does not establish regression against the unmodified-source binary. P1 remains open and auto-merge is paused. Access gap: `problems/native-koblitz-measurement-access.md`.